

Sparse and Deep Gaussian Processes closure for 2-D Fluids and Ocean flows

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Motivation and Challenges

Motivation:

Fully resolved physics:

- High accuracy at High Cost
- Need for reduction

Under-resolved physics:

- Lower accuracy at Lower Cost
- Truncation/discretization error

Challenges:

- Complex nonlinear, multiscale fluid dynamics with unresolved sub-grid physics
- Truncated numerical models introduce truncation and discretization biases
- Regime variability (Reynolds number shifts, Transient forcing) and evolving flow structures

Needs:

- Fast, online data-driven correction of LR dynamics embedded directly into the solver
- Uncertainty-aware predictions of resolved fields and wake metrics
- Transferable across resolutions, domains, and flow conditions

Sparse variational Gaussian Process (SVGP) Closure

Why Gaussian Process?

- Structured priors:** physical smoothness and invariances encoded through Kernel
- Non-parametric flexibility:** expressive function modeling without rigid parametric assumptions
- Uncertainty quantification:** posterior variance enables OOD detection, confidence-based gating of predictions, and natural extension to stochastic solvers

Gaussian Process Regression

$$f(x) \sim \mathcal{GP}(\mu(x), k(x, x'))$$

$$\mu_*(x_*) = k_{*X} K_{XX}^{-1} y, \quad \sigma_*^2(x_*) = k_{**} - k_{*X} K_{XX}^{-1} k_{X*}$$

Multi-Output GP (Output-as-Input view)

$$f(x, i) \sim \mathcal{GP}(\mu(x, i), k((x, i), (x', j))), \quad i \in \{1, \dots, D\}.$$

Linear Model of coregionalization (LMC)

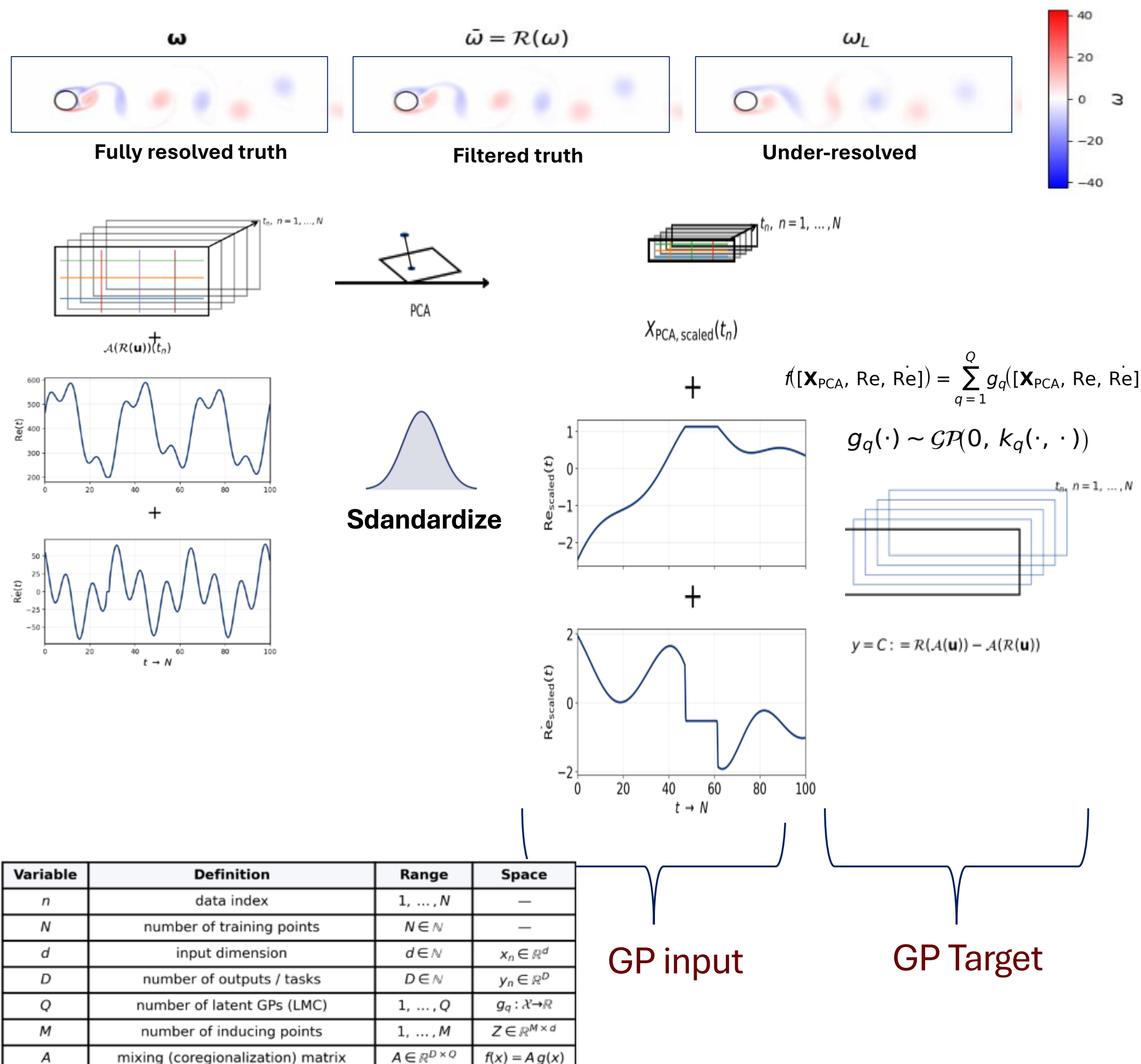
$$g_q(x) \sim \mathcal{GP}(0, k_q(x, x')), \quad q = 1, \dots, Q$$

$$f_p(x) = \sum_{q=1}^Q a_{pq} g_q(x)$$

$$K_f(x, x') = A K_g(x, x') A^\top$$

SVGP Training: Closure definition, Inputs and Outputs

$$\frac{du}{dt} = A(u) + Du + \nabla p$$



SVGP Training: Training Objectives

Model	Kernel	Mean	Loss terms	Effect
M1	$k_{q, \text{RBF}} = \sigma_q^2 \exp\left(-\frac{\ x - x'\ ^2}{2\ell_q^2}\right)$	$b_q + a_q \text{Re}$	ELBO	Baseline
M2	$k_{q, \text{RBF}} = \sigma_q^2 \exp\left(-\frac{1}{2} \sum_{d=1}^D \frac{(x_d - x'_d)^2}{\ell_{q,d}^2}\right)$	$b_q + a_q \text{Re}$	ELBO	ARD sensitivity
M3	$k_q = k_{q, \text{RBF}} + k_{q, \text{Mat}}$ $k_{q, \text{Mat}} = \sigma_q^2 (1 + \sqrt{3} r_q) \exp(-\sqrt{3} r_q)$ $r_q = \left(\sum_{d=1}^D \frac{(x_d - x'_d)^2}{\ell_{q,d}^2}\right)^{1/2}$	$b_q + a_q \text{Re}$	ELBO + $\lambda_{\text{Re}} \ \theta\ _2^2$	Improved amplitude
M5	$k_q = k_{q, \text{RBF}} + k_{q, \text{Mat}}$	$b_q + a_q \text{Re}$	+ $\mathcal{L}_{\text{WNLL}}$	Rare-event correction
M6	$k_q = k_{q, \text{RBF}} + k_{q, \text{Mat}}$	$b_q + a_q \text{Re}$	+ $\mathcal{L}_{\ell, \text{cap}}$	Prevents oversmoothing
M7	$k_q = k_{q, \text{RBF}} + k_{q, \text{Mat}}$	$b_q + a_q \text{Re}$	+ $\beta_{\text{bal}} \mathcal{L}_{\text{bal}}$	Wake-complement balance

Where:

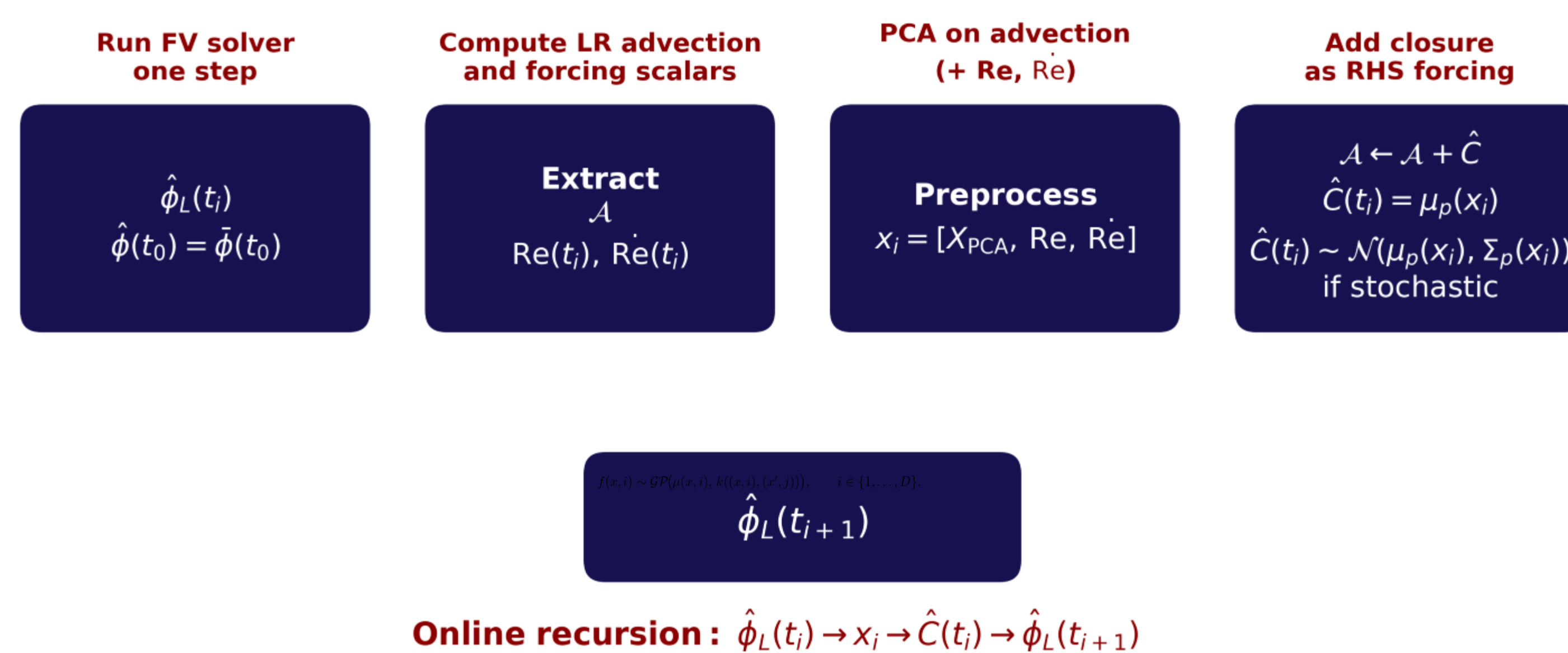
$$\|\theta\|_2^2 = \sum_{q=1}^Q a_q^2 + \sum_{q=1}^Q b_q^2$$

$$\mathcal{L}_{\ell, \text{cap}} = \lambda_{\ell} E_{q, d} [\max(0, \log \ell_{q, d} - \log \ell_{\max})^2]$$

$$\mathcal{L}_{\text{bal}} = (\mathcal{E}(\Omega_{\text{wake}}) - \mathcal{E}(\Omega_{\text{wake}}^c))^2$$

$$\mathcal{L}_{\text{WNLL}} = \frac{1}{N} \sum_{p, n} w_{p, n} \left[\frac{1}{2} \log \sigma_{p, n}^2 + \frac{(y_{p, n} - \mu_{p, n})^2}{2\sigma_{p, n}^2} \right]$$

SVGP Inference: Online Finite Volume GP predictions



Application #1 : In distribution inference, Tidally Modulated Flow Past an Obstacle

Experiment goals

Physics objectives

- Learn transient wake dynamics under modulated $\text{Re}(t)$
- Capture vortex shedding frequency shifts
- Represent acceleration via $d\text{Re}/dt$

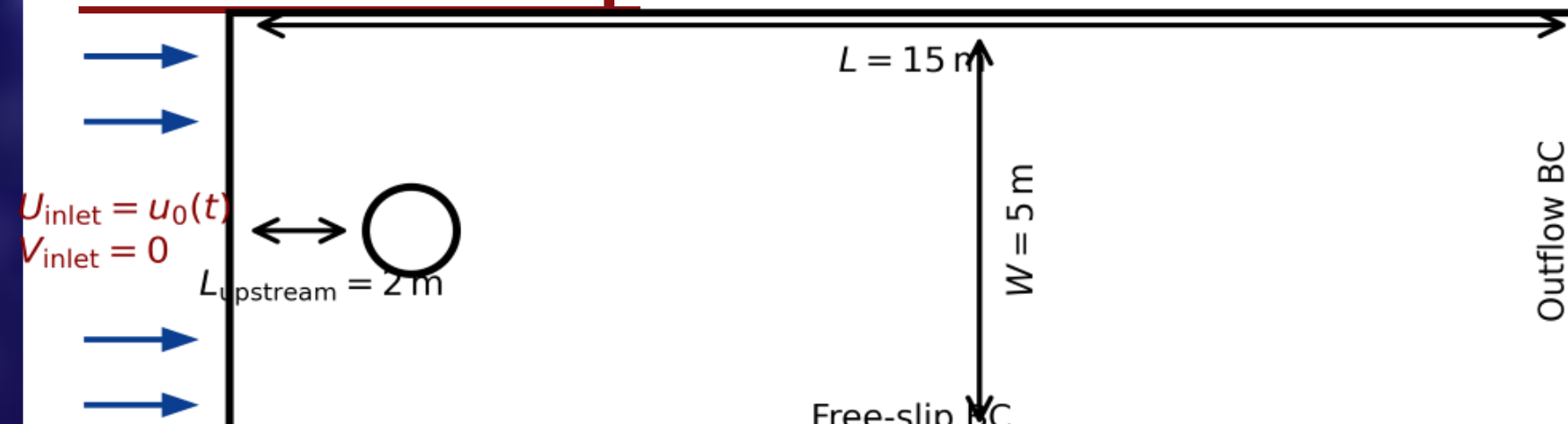
Statistical learning

- Learn time dependent closure statistics
- Maintain accuracy under slow distribution drift
- Generalize beyond training cycles

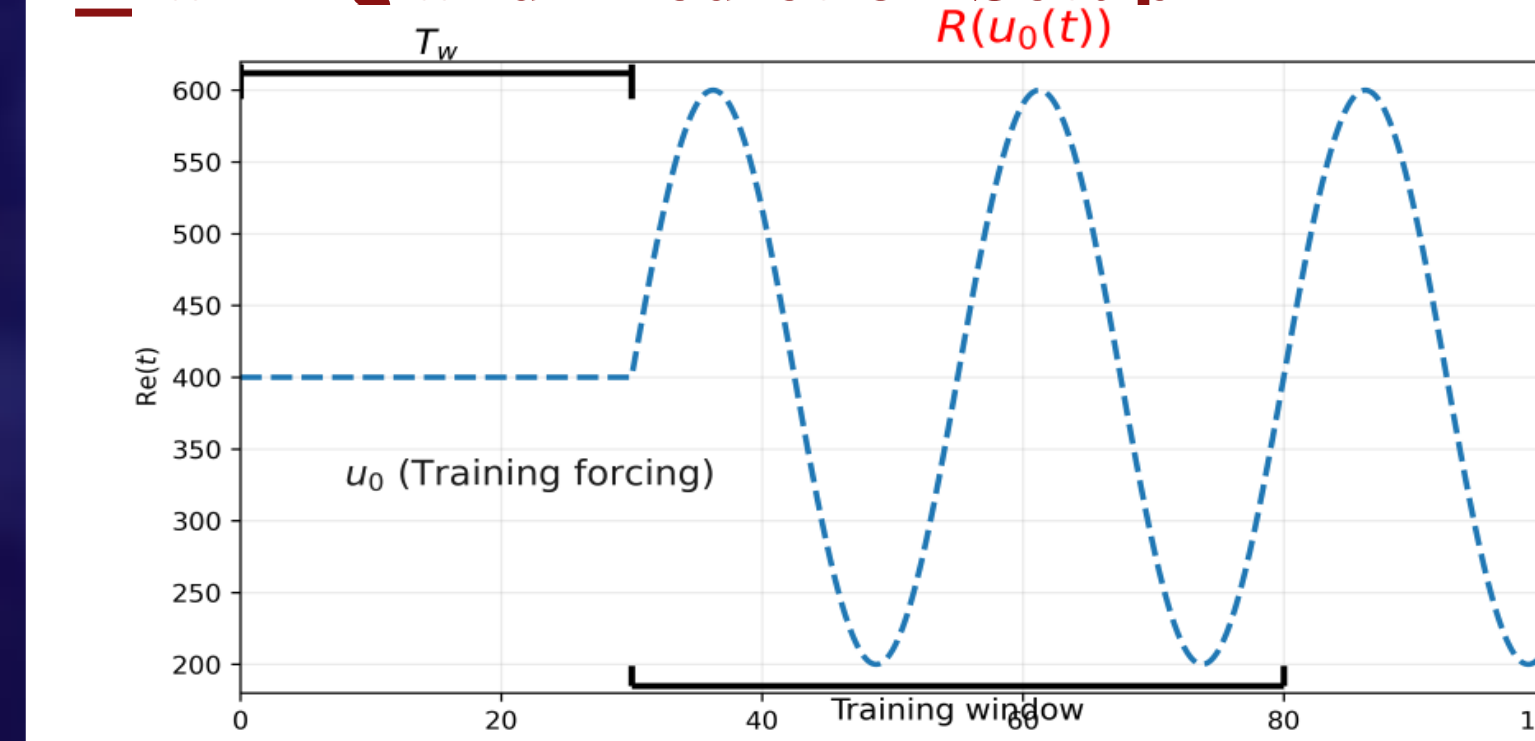
Online deployment

- Stable FV-GP coupling
- Bounded long horizon growth
- Detect OOD via predictive variance
- Robustness under recursive feedback

Simulation Setup

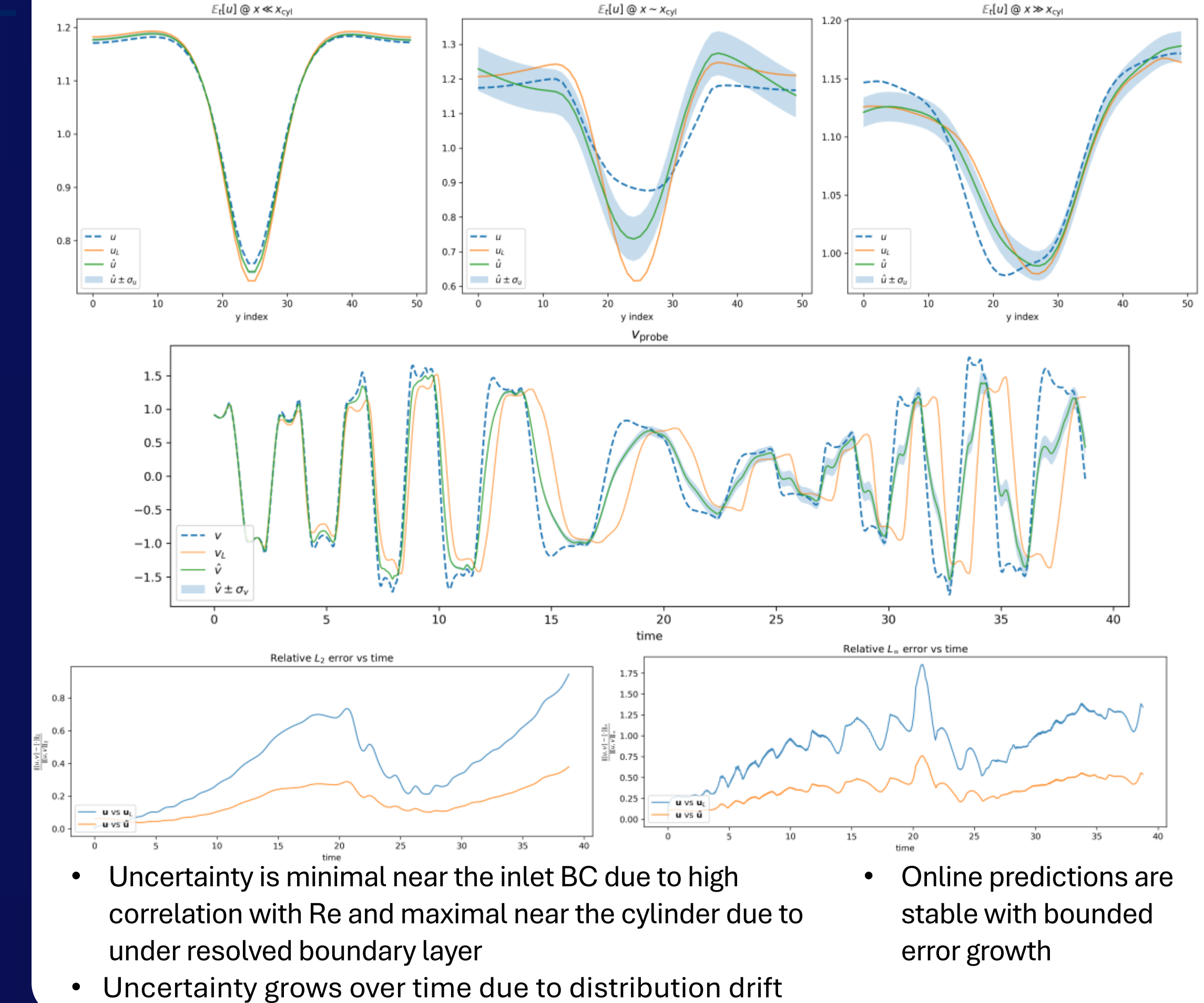


Training and Prediction Setup



- GP is trained on 2 Re modulation cycles which corresponds to ten eddies (events).
- Offline closure evaluations are successful with little to no dependence on the time of the evaluation
- However, in online deployment due to non-linear error accumulation, prediction input samples eventually become OOD as well

Application #1 : Results and discussions

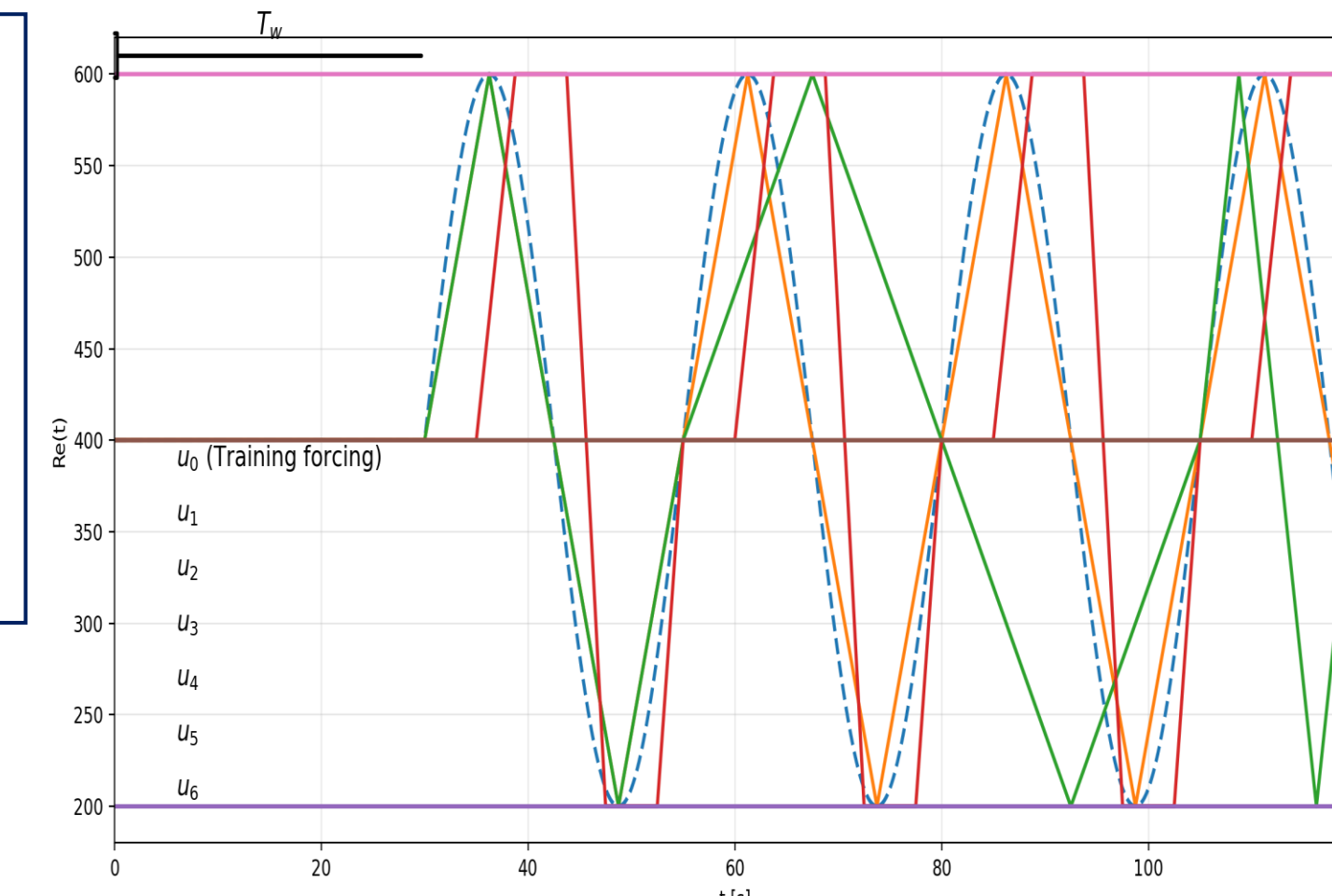


Application #2 : Out of distribution inference, Modulated Flow Past an Obstacle

Experiment goals

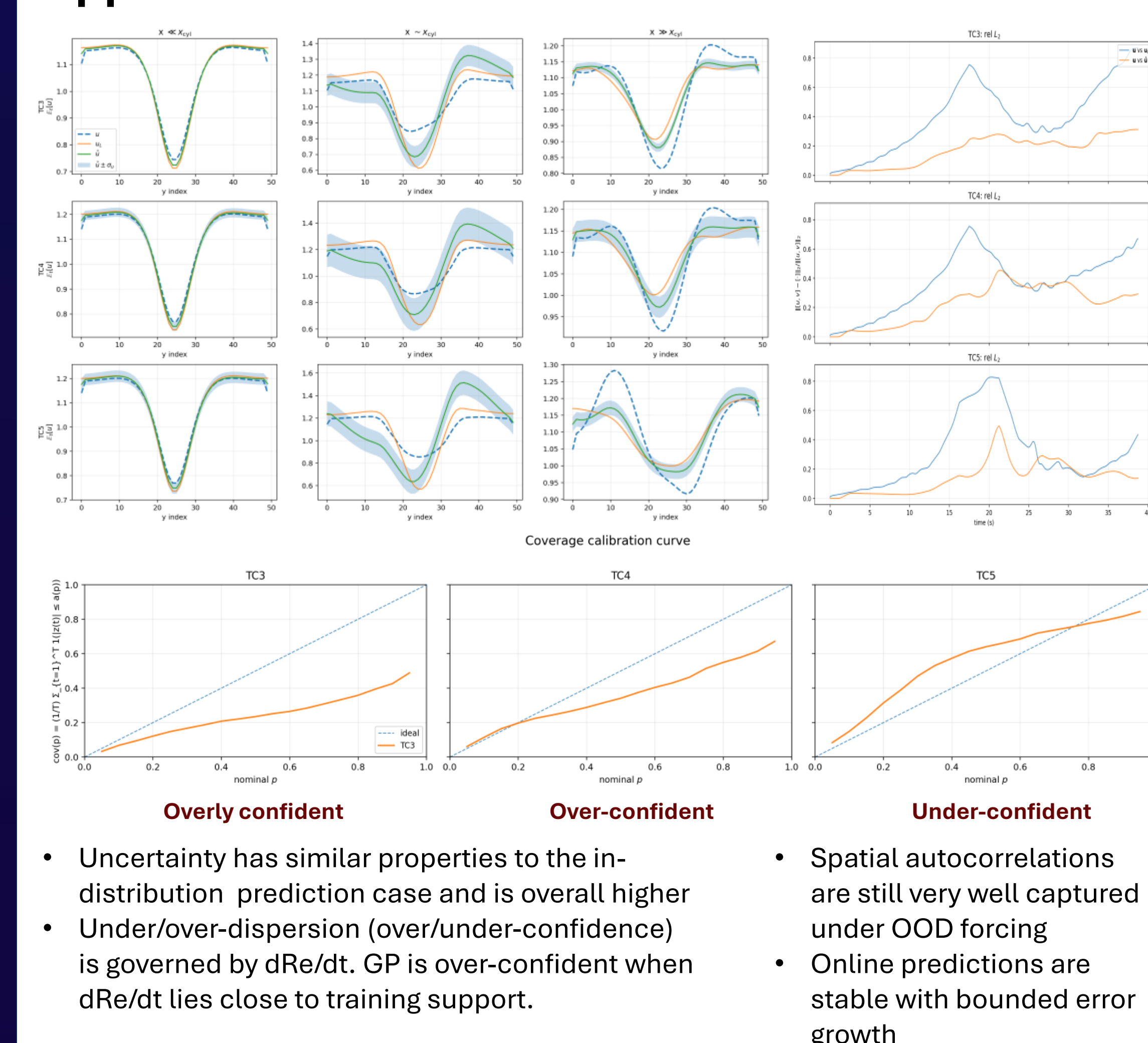
- All the goals from Application #1
- Online FV-GP inference on out of distribution samples
- Extrapolation to unseen physics

OOD forcing functions



Test case	Training forcing	Testing forcing	Training Re range	Testing Re range
TC1	u0	u0	[200,600]	[200,600]
TC2	u0	u0	[400,600]	[200,600]
TC3	u0	u1	[200,600]	[200,600]
TC4	u0	u2	[200,600]	[200,600]
TC5	u0	u3	[200,600]	[200,600]
TC6	u0	u4	[200,600]	[200]
TC7	u0	u5	[200,600]	[400]
TC8	u0	u6	[200,600]	[600]

Application #2 : Results and discussions



Conclusion

We built a sparse variational Gaussian process Finite Volume framework to seamlessly inject a mean GP or stochastic GP term inside the equation and tested it on in- and out of -distribution flows

Acknowledgments